Michael Leong Jin Shun

michaeljs.leong@gmail.com | github.com/Irokanade | irokanade.vercel.app

SKILLS

Systems : Multithreading, Lock-Free Programming, Memory Ordering, Cache Coherence (MESI), Benchmarking, Profiling
Programming Languages : C++, C, Java, Python, TypeScript, SQL
Backend & Frontend : Spring Boot, Ninja Framework, RESTful APIs, Microservices, RabbitMQ, Vue.js
Databases & Tools : Linux, CMake, Git, Docker, Kubernetes, Ansible, MySQL, MongoDB, JFR
Spoken Languages : English (TOEFL iBT: 5.5/6), Mandarin, Bahasa Malaysia

WORK EXPERIENCE

Wisdom Garden Jan 2025 – Present
Full-stack Engineer New Taipei, Taiwan

- Developed and maintained Java (Ninja Framework) REST APIs for a smart space management platform built with AUO, covering meeting-room booking, IoT lighting control, and occupancy detection
- Built event-driven Spring Boot microservices consuming RabbitMQ events to sync bookings to Outlook (Microsoft Graph API) and Google Calendar, and to auto-release no-show bookings from IoT occupancy events
- Built an in-house RabbitMQ channel pool, as the Ninja Framework has no built-in channel caching
- Migrated Jetty 12 to Java virtual threads, raising throughput 15% and cutting peak-load errors from 7.3% to 0% (k6)

Wisdom Garden Jul 2023 – Aug 2023
DevOps Intern New Taipei, Taiwan

- Deployed application releases to school servers and conducted testing of new features
- Updated GitLab deployment configs; ran Ansible deployments with Docker Compose and Kubernetes

PROJECTS

Limit Order Book | github.com/Irokanade/mini-order-book C++20, CMake

- Built a header-only C++20 limit order book that replays NASDAQ add/cancel/execute/replace messages
- Modeled price levels as intrusive FIFO lists with O(1) cancel via hash map, backed by a free-list object pool
- Benchmarked on 600K+ real NASDAQ messages: 18M msgs/sec throughput, 41 ns p50 / 84 ns p99 latency

Lock-Free Single Producer Single Consumer Queue | github.com/Irokanade/iro-spsc-queue C++20

- Implemented a header-only, lock-free SPSC ring buffer in C++20 using acquire/release atomics instead of locks
- Padded indices onto separate cache lines to avoid false sharing; cached the remote index to cut cross-core traffic

CPU Cache Simulator | github.com/Irokanade/iro-cache-simulator C++

- Simulated a 4-core Intel Core i7 cache hierarchy (L1/L2/L3) with the MESI coherence protocol in C++
- Implemented write-back policy, pseudo-LRU replacement, and per-level miss-rate performance counters
- Showed column-major traversal incurs 64x more L1D misses than row-major (16.8M vs. 262K)

Chess Engine and GUI | github.com/Irokanade/iro-chess-gui C, C++, Java

- Implemented a bitboard chess engine in C with alpha-beta, quiescence search, and a Zobrist transposition table
- Built a Java Swing GUI driving the engine over UCI, with a C++ move generator (adapted from Surge) via JNI

ACADEMIC PROJECTS

Teeth Detection and Segmentation System | Senior Project, NTNU 2023 – 2024

- Detected and segmented teeth in dental images with YOLO and SAM, on data labeled with NTU dentists

EDUCATION

National Taiwan Normal University 2020 – 2024
Bachelor of Computer Science and Information Engineering Taipei, Taiwan